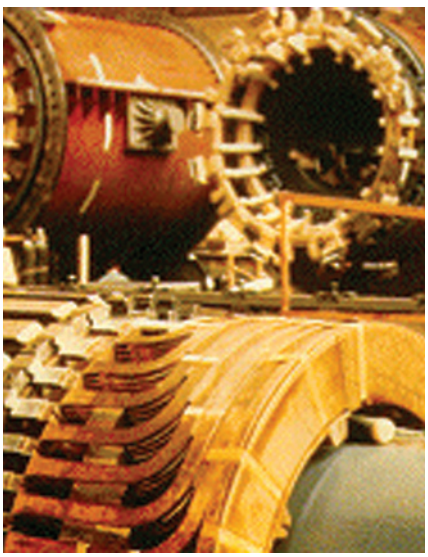
# Chapter 14
# Conductors, insulators and dielectrics





Conductors and insulators. Copper, aluminum and silver (if you can afford it) make good conductors. Pyrex, porcelain and ceramics like alumina make good insulators – they are dielectrics. (Images courtesy of the Copper Development Association.)

**Chapter contents**


| | | |
|---|---|---|
| 14.1 | Introduction and synopsis | 312 |
| 14.2 | Conductors, insulators and dielectrics | 313 |
| 14.3 | Charts for electrical properties | 317 |
| 14.4 | Drilling down: the origins and manipulation of electrical properties | 320 |
| 14.5 | Design: using the electrical properties of materials | 331 |
| 14.6 | Summary and conclusions | 338 |
| 14.7 | Further reading | 338 |
| 14.8 | Exercises | 339 |
| 14.9 | Exploring design with CES | 341 |
| 14.10 | Exploring the science with CES Elements | 343 |

## 14.1 Introduction and synopsis

York Minster, constructed between 1220 and 1400, is one of the great cathedrals of Europe. Many fine sermons have been preached from its pulpit, but when, in 1984, the Archbishop of York expressed his disbelief in the Virgin Birth, the Minster was struck by lightning and severely damaged. Some put this down to the wrath of God, but the reality is simpler: the cathedral was inadequately earthed. There is a message here: even when you are designing cathedrals, electrical properties matter.

This chapter is about the simplest of these: they relate to conduction, insulation and dielectric behavior. *Electrical conduction* (as in lightning conductors) and *insulation* (as in electric plug casings) are familiar properties. Dielectric behavior may be less so. A *dielectric* is an insulator. It is usual to use the word 'insulator' when referring to its inability to conduct electricity, and to use 'dielectric' when referring to its behavior in an electric field. Three properties are of importance here. The first, the *dielectric constant* (or *relative permittivity*), has to do with the way the material acquires a dipole moment (it *polarizes*) in an electric field. The second, the *dielectric loss factor*, measures the energy dissipated when radio-frequency waves pass through a material, the energy appearing as heat (the principle of microwave cooking). The third is the *dielectric breakdown potential*, and this takes us back to York Minster. Lightning is dielectric breakdown, and it can be as damaging on a small scale—in a piece of electrical equipment, for example—as on a large one.

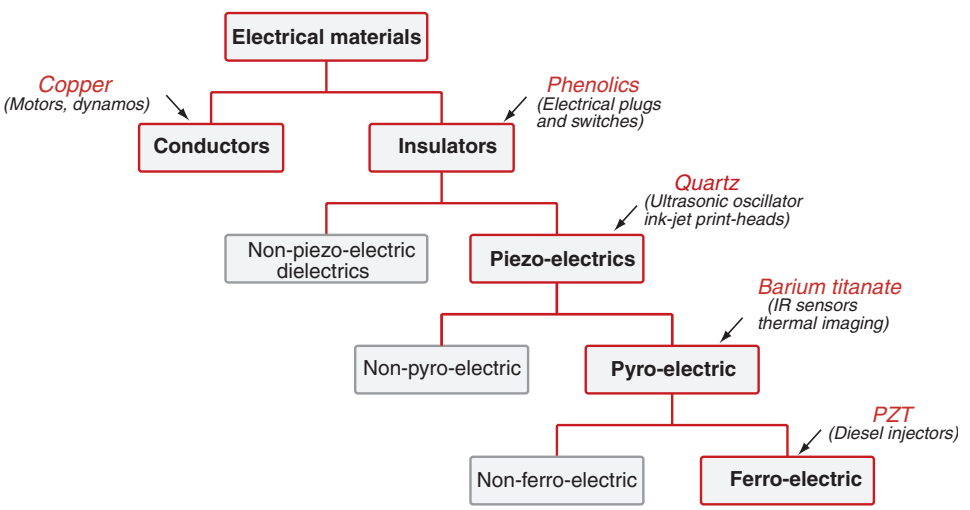



Figure 14.1 The hierarchy of electrical behavior. The interesting ones are boxed in red, with examples of materials and applications. Their nature and origins are described in this chapter.

There are many different kinds of electrical behavior, all of them useful. Figure 14.1 gives an overview, with examples of materials and applications. The chapter introduces electrical properties and their physical origins, and examples of design for electrical performance.

## 14.2 Conductors, insulators and dielectrics

***Resistivity and conductivity***

The electrical resistance $R$ (units: ohms,[1] symbol Ω) of a rod of material is the potential drop $V$ (volts[2]) across it, divided by the current $i$ (amps[3]) passing through it, as in Figure 14.2. This relationship is Ohm's law:

$$R = \frac{V}{i} \tag{14.1}$$

The material property that determines resistance is the *electrical resistivity*, $\rho_e$. It is related to the resistance by

$$\rho_e = \frac{A}{L} R \tag{14.2}$$

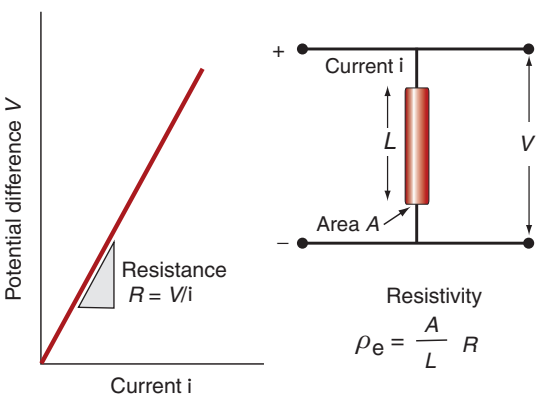



Figure 14.2 Electrical resistivity, $\rho_e$. Its value ranges from 1 to $10^{24}$ μΩ.cm.

[1] George Simon Ohm (1787–1854), dropped out of his studies at the University of Erlangen, preferring billiards to mathematics, but later became absorbed in the theory of electricity and magnetism.

[2] Alessandro Guiseppe Antonio Anastasio Volta (1745–1827), Italian physicist, inventor of the battery or, as it was then known, the Voltaic Pile. His face used to appear on the 10 000 lire note.

[3] André Marie Ampère (1775–1836), mathematician and physicist, who, it was said, had mastered all known mathematics by the age of 14. His life was not a happy one: father guillotined during the French Revolution, miserably married and with difficult children—yet he remained extensively productive up to his death.

where $A$ is the section and $L$ the length of a test rod of material—think of it as the resistance of a unit cube of the material. Its units in the metric system are Ω.m, but it is commonly reported in units of μΩ.cm. It has an immense range, from a little more than $10^{-8}$ in units of Ω.m for good conductors (equivalent to 1 μΩ.cm, which is why these units are still used) to more than $10^{16}$Ω.m ($10^{24}$ μΩ.cm) for the best insulators. The electrical conductivity $\kappa_e$ is simply the reciprocal of the resistivity. Its units are siemens per meter S/m or $(\Omega.\text{m})^{-1}$).

***Dielectric properties***

First, a reminder of what is meant by a field: it is a region of space in which objects experience forces if they have the right properties. Charge creates an *electric field*, $E$. The electric field strength between two oppositely charged plates separated by a distance $t$ and with a potential difference $V$ between them is

$$E = \frac{V}{t} \tag{14.3}$$

and is independent of position except near the edge of the plates.

Two conducting plates separated by a dielectric make a capacitor (Figure 14.3). Capacitors (sometimes called condensers) store charge. The charge $Q$ (coulombs[4]) is directly proportional to the potential difference between the plates, $V$ (volts):

$$Q = CV \tag{14.4}$$

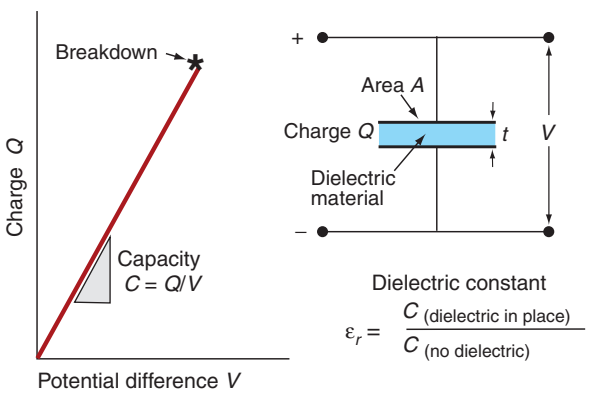



Figure 14.3 Dielectric constant and dielectric breakdown. The capacitance of a condenser is proportional to the dielectric constant, and the maximum charge it can hold is limited by breakdown.

[4] Charles Augustin Coulomb (1736–1806), military physicist, laid the foundations of both the mathematical description of electrostatic forces and the laws of friction. Despite his appointment as *Intendant des Eaux et Fontaines de Paris*, he made no known contributions to hydrodynamics.

where $C$ (farads[5]) is the capacitance. The capacitance of a parallel plate capacitor of area $A$, separated by empty space (or by air), is

$$C = \varepsilon_o \frac{A}{t} \tag{14.5}$$

where $\varepsilon_o$ is the *permittivity* of *free* space ($8.85 \times 10^{-12}$ F/m, where F is farads). If the empty space is replaced by a dielectric, capacitance increases. This is because the dielectric *polarizes*. The field created by the polarization opposes the field $E$, reducing the voltage difference $V$ needed to support the charge. Thus, the capacity of the condenser is increased to the new value

$$C = \varepsilon \frac{A}{t} \tag{14.6}$$

where $\varepsilon$ is the *permittivity of the dielectric* with the same units as $\varepsilon_o$. It is usual to cite, not this, but the *relative permittivity* or *dielectric constant*, $\varepsilon_r$:

$$\varepsilon_r = \frac{C_{\text{with dielectric}}}{C_{\text{no dielectric}}} = \frac{\varepsilon}{\varepsilon_o} \tag{14.7}$$

making the capacitance

$$C = \varepsilon_r \varepsilon_o \frac{A}{t} \tag{14.8}$$

Being a ratio, $\varepsilon_r$ is dimensionless. Its value for empty space and, for practical purposes, for most gases, is 1. Most dielectrics have values between 2 and 20, though low-density foams approach the value 1 because they are largely air. Ferroelectrics are special: they have values of $\varepsilon_r$ as high as 20 000. More on them later.

Capacitance is one way to measure the dielectric constant of a material (Figure 14.3). The charge stored in the capacitor is measured by integrating the current that flows into it as the potential difference $V$ is increased. The ratio $Q/V$ is the capacitance. The dielectric constant $\varepsilon_r$ is calculated from equation (14.7).

Small capacitors, with capacitances measured in microfarads (μF) or picofarads (pF), are used in R–C circuits to tune oscillations and give controlled time delays. The time constant for charging or discharging a capacitor in series with a resistor is

$$\tau = RC \tag{14.9}$$

[5] Michael Faraday (1791–1867), brilliant experimentalist both in physics and chemistry, discoverer of electromagnetic induction and inspiring lecturer at the Royal Institution, London.

where $R$ is the resistance of the circuit. When charged, the energy stored in a capacitor is

$$\frac{1}{2}QV = \frac{1}{2}CV^2 \tag{14.10}$$

and this can be large: 'super-capacitors' with capacitances measured in farads store enough energy to power a hybrid car.

The *breakdown potential* or *dielectric strength* (units: MV/m) is the electrical potential gradient at which an insulator breaks down and a damaging surge of current flows through it. It is measured by increasing, at a uniform rate, a 60 Hz alternating potential applied across the faces of a plate of the material in a configuration like that of Figure 14.3 until breakdown occurs, typically at a potential gradient of between 1 and 100 MV/m.

The *loss tangent* and the *loss factor* take a little more explanation. We shall see in Section 14.4 that polarization involves the small displacement of charge (either of electrons or of ions) or of molecules that carry a dipole moment when an electric field is applied to the material. An oscillating field drives the charge between two alternative configurations. This charge motion is like an electric current that—if there were no losses—would be 90° out of phase with the voltage. In real dielectrics this current dissipates energy, just as a current in a resistor does, giving it a small phase shift, $\delta$ (Figure 14.4). The *loss tangent*, tan $\delta$, also called the *dissipation factor*, $D$, is the tangent of the loss angle. The *power factor*, $P_f$, is the sine of the loss angle. When $\delta$ is small, as it is for the materials of interest here, all three are essentially equivalent:

$$P_f \approx D \approx \tan\delta \approx \sin\delta \tag{14.11}$$

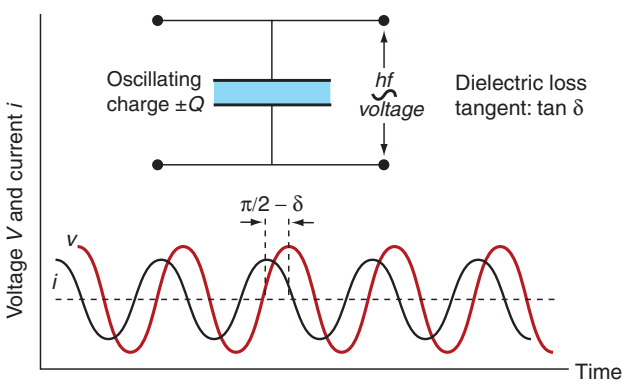



Figure 14.4 Dielectric loss. The greater the loss factor, the greater is the microwave coupling and heating.

More useful, for our purposes, is the *loss factor* $L$, which is the loss tangent times the dielectric constant:

$$L = \varepsilon_r \tan \delta \tag{14.12}$$

It measures the energy dissipated in a dielectric when in an oscillating field. If you want to select materials to minimize or maximize dielectric loss, then the measure you want is $L$.

When a dielectric material is placed in a cyclic electric field of amplitude $E$ and frequency $f$, power $P$ is dissipated and the field is correspondingly attenuated. The power dissipated per unit volume, in W/m$^3$, is

$$P \approx f E^2 \varepsilon \tan \delta = f E^2 \varepsilon_o \varepsilon_r \tan \delta = f E^2 \varepsilon_o L \tag{14.13}$$

where, as before, $\varepsilon_r$ is the dielectric constant of the material and $\tan \delta$ is its loss tangent. This power appears as heat and is generated uniformly through the volume of the material. Thus, the higher the frequency or the field strength and the greater the loss factor $\varepsilon_r \tan \delta$, the greater is the heating and energy loss. This dielectric loss is exploited in processing—for example, in radio-frequency welding of polymers.

All dielectrics change shape in an electric field, a consequence of the small shift in charge that allows them to polarize; the effect is called *electrostriction*. Electrostriction is a one-sided relationship in that an electric field causes deformation, but deformation does not produce an electric field. *Piezo-electric* materials, by contrast, display a two-sided relationship between polarization and deformation: a field induces deformation and deformation induces charge differences between its surfaces, thus creating a field. The piezo-electric coefficient is the strain per unit of electric field, and although it is very small, it is a true linear effect, and this makes it useful: when you want to position or move a probe with nanoscale precision it is just what you need. *Pyro-electric* materials contain molecules with permanent dipole moments that, in a single crystal, are aligned, giving the crystal a permanent polarization. When the temperature is changed the polarization changes, creating surface charges or, if the surfaces are connected electrically, a pyro-electric current—the principle of intruder detection systems and of thermal imaging. *Ferro-electric* materials, too, have a natural dipole moment—they are polarized to start with, and the individual polarized molecules line up so that their dipole moments are parallel, like magnetic moments in a magnet. Their special feature is that the direction of polarization can be changed by applying an electric field, and the change causes a change of shape. Applications of all of these are described in Section 14.5.

## 14.3 Charts for electrical properties

All materials have an electrical resistivity and all dielectrics have a dielectric constant and loss factor. Property charts for these are useful because they allow

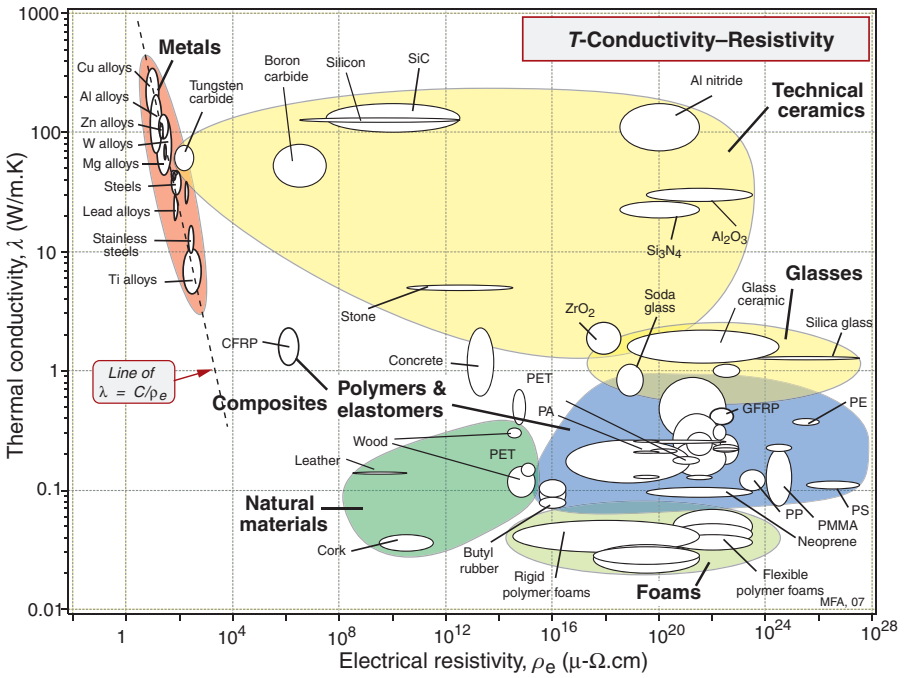



Figure 14.5 Electrical resistivity $\rho_e$ and thermal conductivity $\lambda$.

comparison and selection from a large and varied population. Certain other properties are peculiar to only a handful of materials: useful piezo-electric and ferro-electric behavior are examples. Selection is then a case of a choice between a small number of candidates, easily explored by direct comparison. Property charts for these are not necessary.

***Thermal conductivity and electrical resistivity***

The first chart (Figure 14.5) shows the thermal conductivity $\lambda$ and the electrical resistivity $\rho_e$. The first has a range of $10^5$, the second a range of $10^{28}$; no other material property has such a wide range. The chart is useful for selecting materials for applications in which these two properties are important. For metals, thermal and electrical conduction are linked because both depend on free electrons, giving the obvious correlation between the two at the upper left of the chart:

$$\lambda \approx \frac{1400}{\rho_e}$$

($\lambda$ in W/m.K, $\rho_e$ in $\mu\Omega$.cm). It is known as the Wiedemann–Franz law.